

AMA - GPU (TD 2)

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Reminder

- Do not forget to test for errors, for instance,
`err = clu_Queue->enqueueNDRangeKernel( ... );`
`cluCheckError(err, "kernel");`

Exercise 1: Parallel scan

Implement the parallel scan algorithm presented during the class (see slides). To verify the correctness, compare the result of a baseline computation in the CPU with the obtained result using OpenCL.

Exercise 2: Parallel closest pair of points

Implement in OpenCL a parallel algorithm that, for n points in the 2D Euclidean space, finds the pair of points with the smallest distance between them (see Figure 1). Let us recall that a 2D point p can be expressed by two coordinates p_x and p_y (Cartesian coordinate system) and that the Euclidean distance between two points p and q is defined as $\sqrt{(p_x - q_x)^2 + (p_y - q_y)^2}$. For simplicity, integers represent the coordinates of the points.

First, implement a CPU version to compute the closest distance. Second, implement the OpenCL version and compare the result with the CPU version.

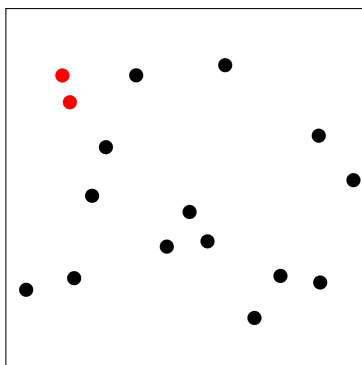


Figure 1: The algorithm finds the pair of points (colored in red) with the smallest distance between them.